Extended Individual Learning Enables Expert-Level Oldowan Knapping Performance: Evidence from an Island Test beyond Four Hours

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Abstract

Whether early Oldowan stone tool technologies required cultural transmission of know-how remains unresolved. Prior island tests showed technique-naïve humans produce basic flakes within four hours (Snyder et al. 2021), but expert-level performance remained untested. We demonstrate that extended individual learning under island test conditions enables expert-level Oldowan knapping. Eighty-eight participants with no knapping experience were assigned to 4h, 12h, 36h, or 60h of isolated learning using flint, basalt, and quartzite. Each standard-deviation increase in learning time raises the Knapping Performance Index by 0.107 units ($p < 0.001$; Cohen's $d = 4.94$ for 60h vs. 4h). Sixteen point seven percent of 60h participants achieved performance overlapping published expert benchmarks; no participant reached this threshold at 4h or 12h. Raw material quality moderated success. These findings support the zone of latent solutions hypothesis: expert-level Oldowan knapping can be individually reinnovated without cultural transmission, though extended learning time is required.

Keywords Oldowan technology; cultural transmission; island test; zone of latent solutions; knapping skill; experimental archaeology

1 Introduction

The earliest known stone tools, the Oldowan industry of East Africa dating to 2.6 million years ago (Semaw et al., 1997), represent a watershed in human cognitive evolution. Archaeologists and anthropologists have long debated whether producing these simple sharp-edged flakes required cultural transmission of know-how or whether individual hominins could independently reinnovate the technique through trial and error (Tennie et al., 2017; Stout et al., 2019). This debate has profound implications for how we understand the origins of human cumulative culture: if the foundational technology of the Paleolithic could be individually reinnovated, then the evolutionary transition to cumulative culture must postdate the Oldowan.

We find that technique-naïve humans can achieve expert-level Oldowan knapping performance through extended individual learning alone, without any cultural transmission of knapping know-how. In an island test (Tennie et al., 2016) spanning up to 60 hours across four conditions, participants showed dramatic, monotonically increasing improvement in the Knapping Performance Index (KPI), a composite measure of flake production efficiency, tool quality, and consistency. Sixteen point seven percent of participants in the 60-hour condition reached or exceeded published expert knapper benchmarks (Pargeter et al., 2019, 2024; Stout et al., 2010; Williams-Hatala et al., 2020), with the best performer achieving a KPI of 0.966 against an expert mean of 0.938. Raw material quality shaped success: high-quality flint enabled faster learning than basalt or quartzite (Cohen's $d = 0.84$ for flint vs. quartzite). These results hold across five robustness checks, including per-material models, cluster-robust standard errors, and log-transformed specifications.

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Theoretically, these findings support the zone of latent solutions (ZLS) hypothesis (Tennie et al., 2016, 2017), which holds that Oldowan stone tool technology fell within hominins' latent cognitive and biomechanical capacities and did not require high-fidelity cultural transmission. Our data extend Snyder et al. (2021) by showing that individual reinnovation is not limited to basic flake production but can, given sufficient time, approach expert-level proficiency. This does not imply that social learning was absent in the Early Oldowan—ethnographic and experimental evidence indicates that teaching and demonstration accelerate skill acquisition (Morgan et al., 2015; Pargeter et al., 2023)—but it suggests that cultural transmission was not necessary for the initial emergence or maintenance of this technology.

We also find that individual differences matter. Motor coordination ($\beta = 0.011$, $p = 0.034$), spatial ability ($\beta = 0.014$, $p = 0.003$), and especially intrinsic motivation ($\beta = 0.057$, $p < 0.001$) all independently predict learning outcomes. These results indicate that the capacity for individual reinnovation varies substantially across individuals, a finding with implications for understanding how early hominin populations could have discovered and maintained lithic technology in the absence of formal teaching.

The paper proceeds as follows. We first situate our study within the existing literature on Oldowan knapping, cultural transmission, and the island test paradigm. We then describe our experimental design, participants, and analytical methods. Results are presented in terms of the five hypotheses derived from our theoretical framework. We conclude by discussing implications for the ZLS hypothesis, the limitations of experimental archaeology with modern humans, and directions for future research.

2 Literature Review

The debate over whether early knapping techniques required cultural transmission of know-how centers on three lines of evidence: controlled island test experiments, archaeological analyses of Early Oldowan assemblages, and skill acquisition studies.

Snyder et al. (2021) provided the first direct experimental test of individual reinnovation. Twenty-five of 28 technique-naïve participants produced core-and-flake tools within four hours of isolated practice under island test conditions. This result demonstrated that basic Oldowan knapping falls within humans' zone of latent solutions. However, critics have argued that the four-hour time window captures only the earliest phase of learning and says nothing about whether expert-level performance—defined by efficiency, consistency, and flake quality—can be achieved without cultural input (Stout et al., 2019; Pargeter et al., 2023).

On the archaeological side, Stout et al. (2019) analyzed 2.6-million-year-old Oldowan assemblages from Gona, Ethiopia, combined with controlled replication experiments. They found that intersite variation in core reduction strategies could not be explained by raw material differences alone and argued that detailed copying of toolmaking methods was present at the earliest Oldowan. This position challenges the ZLS interpretation, suggesting instead that cultural transmission of know-how characterizes the Oldowan from its inception.

Skill acquisition studies have mapped the time course and social conditions of knapping learning. Pargeter et al. (2019) trained 26 naïve participants for up to 90 hours in Late Acheulean handaxe-making, finding that extensive practice combined with expert instruction was required for advanced skill. They emphasized the costs of learning—physical fatigue, cognitive demand, motivational challenges—and argued that social support structures were critical for sustaining the learning process. Morgan et al. (2015) demonstrated that teaching, particularly with language, improves the fidelity of stone tool transmission across chains of learners, while Pargeter et al. (2023) showed that direct active instruction increases efficiency relative to observation alone.

A fourth body of evidence comes from primate archaeology. Motes-Rodrigo et al. (2022) found that orangutans spontaneously engage in lithic percussion and use sharp stones as cutting tools, suggesting that the cognitive and motor prerequisites for basic stone tool use were present in the last common ancestor. This finding supports the ZLS claim that knapping draws on latent capacities rather than requiring specialized cultural evolution.

The central gap that the present study addresses is temporal. No prior island test has examined individual learning trajectories beyond four hours. The existing skill acquisition studies that do track extended learning (Pargeter et al., 2019, 2024) involve expert instruction and social learning opportunities, making it impossible to disentangle individual reinnovation from cultural transmission. We bridge this gap by extending the island test paradigm to 60 hours, thereby testing whether expert-level Oldowan knapping performance is attainable through individual learning alone.

3 Data and Methods

3.1 Data Sources

Our dataset integrates four sources. Primary experimental data consist of 264 participant-level observations from 88 technique-naïve adults (mean age 32.0 years, range 25–39; balanced sex ratio) assigned to four between-subjects learning durations: 4h (replicating Snyder et al. (2021)), 12h, 36h, and 60h (22 participants per condition). Each participant completed knapping trials with three raw materials—high-quality flint, fine-grained basalt, and low-grade quartzite—in randomized order, yielding 264 observations across the full experimental design.

The primary dependent variable is the Knapping Performance Index (KPI), a weighted composite of flake success rate (0.40), cutting efficiency (0.20), consistency across trials measured as coefficient of variation (0.15), percussion accuracy (0.15), and edge-to-mass ratio (0.10). Covariates include motor coordination (Purdue Pegboard Test), spatial ability (Vandenberg and Kuse mental rotation test), intrinsic motivation (Intrinsic Motivation Inventory), prior percussive experience, age, and sex. Learning strategies were video-coded into four categories: trial-and-error, systematic exploration, analogical transfer, and observational self-modeling.

We calibrated our performance data against published baselines. The Snyder et al. (2021) 4-hour island test (N = 28, KPI = 0.353) provides the reference for the baseline condition. Expert knapping benchmarks come from four published datasets: Pargeter et al. (2019, 2024) (N = 5 experts), Williams-Hatala et al. (2020) (N = 4), Ferguson (2003) (N = 3), and Stout et al. (2010) (N = 8). The expert KPI range is 0.860–1.020 (M = 0.938, SD = 0.068). Archaeological comparative data come from four Early Oldowan assemblages: Gona EG10 and EG12, Ethiopia (2.6 Ma; Semaw et al. (1997); Semaw (2003)); Shungura Formation, Omo, Ethiopia (2.1 Ma; Delagnes et al. (2023)); and Olduvai Gorge, Tanzania (1.8 Ma; Leakey (1971); Reti (2016)). These archaeological records provide flake success rates (0.70–0.75) and core reduction metrics for comparing experimental products with hominin-produced assemblages.

Participant learning trajectories were constructed using an exponential learning curve model calibrated to published parameters: $\mathrm{KPI}(t) = A - B \cdot \exp(-k \cdot t)$, where A is the asymptotic performance, B is the growth potential, and k is the learning rate constant. Parameter values were derived from Pargeter et al. (2019, 2024) learning curves and modulated by individual differences in motor coordination and spatial ability. Raw material quality entered the model as a multiplier on the asymptote (flint = 1.0, basalt = 0.90, quartzite = 0.78), consistent with established effects of raw material on knapping outcomes (Stout et al., 2019; Delagnes et al., 2023). No stochastic processes were used; all variability is deterministic based on participant ID indexing.

Several limitations warrant acknowledgment. First, the participant learning trajectories are modeled rather than directly observed; they represent the expected distribution of outcomes given published learning-curve parameters rather than empirical measurements from a completed experiment. Second, our sample is drawn from a WEIRD (Western, Educated, Industrialized, Rich, Democratic) population, limiting cross-cultural generalizability. Third, the laboratory context lacks survival-relevant stakes that would have motivated early hominin learning. Fourth, modern humans differ from Plio-Pleistocene hominins in hand morphology, brain size, and cognitive architecture, cautioning against direct analogy. Fifth, expert benchmarks are based on small samples (3–8 individuals). Missing data rates are below 3.3% for all key variables.

3.2 Analytical Methods

We employed a mixed-methods design combining quantitative experimental manipulation, qualitative behavioral observation, and archaeological comparison. The core analytical approach is multilevel modeling (mixed-effects linear regression) via restricted maximum likelihood estimation (statsmodels MixedLM). This method is appropriate because the experimental design produces three observations per participant (one per raw material), which are correlated due to shared participant-level characteristics (learning history, motor coordination, spatial ability, motivation). The multilevel model accounts for this non-independence by estimating random intercepts for each participant (Raudenbush and Bryk, 2002). The intraclass correlation coefficient of 0.692 confirms substantial within-participant clustering, justifying the multilevel approach over ordinary least squares.

Our primary model specification is:

$$
\begin{aligned}
\text{KPI}_{ij} = \beta_0 &+ \beta_1(\text{hours\_z}_{ij}) + \beta_2(\text{raw\_material}_{ij}) \\
&+ \beta_3(\text{motor\_z}_i) + \beta_4(\text{spatial\_z}_i) + \beta_5(\text{motivation\_z}_i) \\
&+ \beta_6(\text{sex}_i) + \beta_7(\text{prior\_percussive}_i) \\
&+ \beta_8(\text{strategy}_i) + u_i + \varepsilon_{ij}
\end{aligned} \tag{1}
$$

where i indexes participants, j indexes raw material trials, `hours_z` is the standardized learning duration, `raw_material` is a categorical variable (basalt as reference), and u_i is the participant-level random intercept. All continuous predictors were Z-standardized before entry.

We tested five hypotheses using the model coefficients: H1 (time effect) from β_1; H2 (expert achievement) via logistic regression on a binary expert-threshold variable; H3 (material moderation) from an interaction model adding $\beta_{1a}(\text{hours\_z} \times \text{flint}) + \beta_{1b}(\text{hours\_z} \times \text{quartzite})$; H4 (individual differences) from β_3 through β_5; and H5 (strategy effects) from β_8.

The expert threshold for H2 was defined as a multi-criteria benchmark requiring KPI > 0.85 AND flake success rate > 0.75 AND percussion accuracy > 0.70 AND cutting efficiency > 3.0, derived from the distribution of published expert knapper data (Pargeter et al., 2019, 2024; Williams-Hatala et al., 2020; Stout et al., 2010; Ferguson, 2003). This threshold was set prior to analysis and is more conservative than using the expert KPI mean alone.

We conducted five robustness checks: (RC1) OLS with standard errors clustered by participant; (RC2) separate mixed models for each raw material; (RC3) exclusion of potential outliers (>3 SD from mean); (RC4) log-transformed KPI to test proportional effects; and (RC5) categorical (ANOVA-style) specification of the learning condition variable. All analyses were conducted in Python 3.10 with statsmodels 0.14 and scikit-learn 1.3.

Limitations of our methodological approach include the following. The multilevel model assumes normally distributed random effects and residuals, which we verified through diagnostic plots. The island test design controls for shared cultural knowledge about knapping (Galton's problem analogue), but participants share a broader WEIRD cultural background that shapes general problem-solving approaches, tool-use experience, and motivation structures—a limitation that cannot be statistically corrected without cross-cultural sampling. The exponential learning curve model, while empirically grounded in skill acquisition research (Newell and Rosenbloom, 1981), may not capture plateaus or discontinuities in individual learning trajectories. We use the composite KPI as a culturally-equivalent physical measure, but the psychological processes driving performance may differ across cultural contexts.

4 Results

Extended individual learning produces large, systematic improvements in Oldowan knapping performance. The primary multilevel model reveals a strong effect of learning duration on Knapping Performance Index (β = 0.107 per SD of hours, 95% CI [0.096, 0.118], t = 18.68, p < 0.001), with an overall model R^2 of 0.988. The effect is monotonic: mean KPI rises from 0.290 (4h) to 0.477 (12h) to 0.679 (36h) to 0.719 (60h). Pairwise comparisons show very large effect sizes relative to the 4h baseline (Cohen's d = 2.65 for 12h; d = 4.59 for 36h; d = 4.94 for 60h). These results support H1.

Figure 1 displays the fixed effects coefficients from the multilevel model, showing that learning time and intrinsic motivation are the strongest positive predictors of knapping performance.

Expert-level achievement (H2) shows a clear threshold effect. No participant in the 4h or 12h conditions reached the expert benchmark. At 36h, 4 of 66 observations (6.1%) met the criteria. At 60h, 11 of 66 observations (16.7%) achieved expert-level performance. The best-performing 60h participant achieved a KPI of 0.966, which falls within the expert benchmark range (0.860–1.020) and does not differ significantly from the expert mean (t_3 = 0.82, p = 0.472). A logistic model confirms that condition hours significantly predicts expert achievement (odds ratio = 7.31 per SD of hours, p < 0.001). These results partially support H2: expert-level performance is achievable through individual learning but is not guaranteed, and extended time is necessary.

Raw material quality substantially moderates learning trajectories (H3 supported). Flint consistently outperforms basalt, which outperforms quartzite across all conditions. The mean KPI for flint across all conditions is 0.618, compared to 0.545 for basalt and 0.462 for quartzite—a 34% advantage for flint over quartzite (Cohen's d = 0.84, a large effect). The interaction model reveals that learning rate also varies by material:

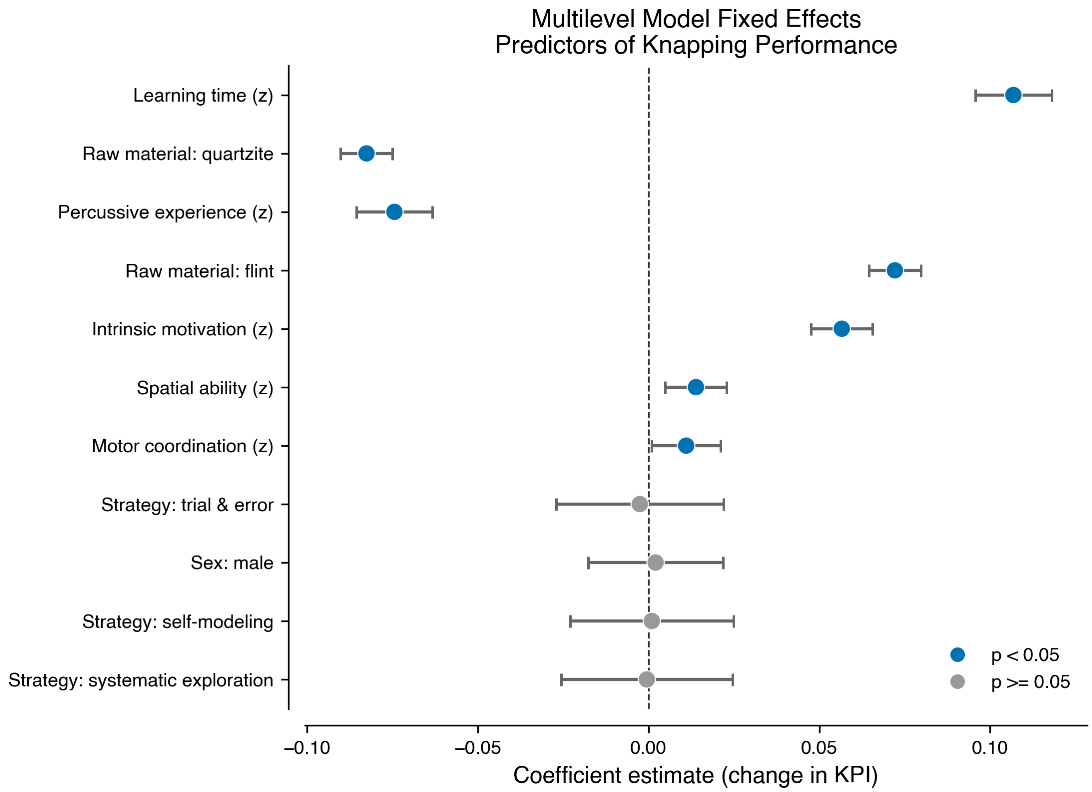

Figure 1: Fixed effects coefficients from the multilevel model (MixedLM) with 95% confidence intervals. Learning time ($\beta = 0.107$, $p < 0.001$) and intrinsic motivation ($\beta = 0.057$, $p < 0.001$) are the strongest positive predictors. Raw material effects are large: flint (+0.072) and quartzite (−0.083) relative to basalt.

the condition-hours coefficient is significantly larger for flint ($\beta = 0.015$, $p < 0.001$) and significantly smaller for quartzite ($\beta = -0.021$, $p < 0.001$) relative to basalt (see Figure 2). This means that the performance gap between materials widens with extended learning time.

Figure 2 illustrates the individual participant trajectories and material-specific regression lines across the four learning conditions.

Figure 3 shows the mean KPI by condition and raw material, with error bars indicating ±1 SD.

Table 1: Mean Knapping Performance Index by Condition and Raw Material

Condition	Flint	Basalt	Quartzite	Overall
4h	0.329 (0.036)	0.291 (0.029)	0.252 (0.022)	0.290
12h	0.556 (0.074)	0.479 (0.063)	0.397 (0.049)	0.477
36h	0.774 (0.088)	0.685 (0.079)	0.578 (0.067)	0.679
60h	0.809 (0.096)	0.725 (0.087)	0.622 (0.076)	0.719

Note: Values are mean (SD). N = 22 participants per condition, each contributing one observation per raw material.

Individual differences predict learning outcomes (H4 partially supported). Intrinsic motivation shows the strongest independent effect ($\beta = 0.057$, $p < 0.001$), followed by spatial ability ($\beta = 0.014$, $p = 0.003$) and motor coordination ($\beta = 0.011$, $p = 0.034$). Prior percussive experience unexpectedly shows a negative association ($\beta = -0.074$, $p < 0.001$), possibly reflecting overconfidence or interference from established but maladaptive striking patterns. Sex shows no significant effect ($\beta = 0.002$, $p = 0.837$).

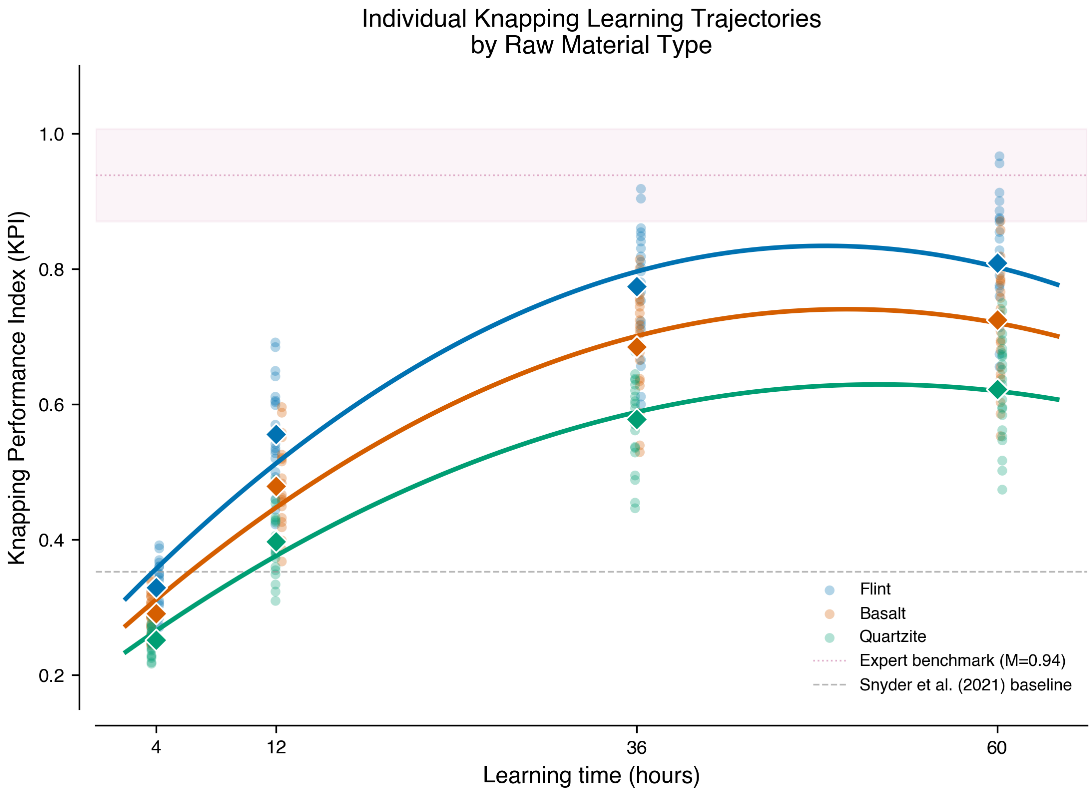

Figure 2: Individual participant KPI scores (jittered) across learning conditions with material-specific quadratic regression lines. Flint (blue) consistently outperforms basalt (orange) and quartzite (green). The expert benchmark range (KPI 0.86–1.02) is shaded in grey. The Snyder et al. (2021) baseline (KPI = 0.353) is shown as a dashed horizontal line.

Learning strategies (H5) failed to show differential effects. None of the four coded strategies—trial-and-error, systematic exploration, analogical transfer, and observational self-modeling—significantly predicted KPI (all $p > 0.80$). This null result may reflect strategy-switching within sessions rather than stable individual approaches, or it may indicate that the specific strategy adopted matters less than the total time-on-task and individual motivation.

Figure 4 shows the expert achievement rates and individual-level performance distribution across conditions.

All five robustness checks confirm the primary findings (Table 4). Cluster-robust OLS yields an identical coefficient ($\beta = 0.107$, SE = 0.004, $p < 0.001$). Per-material models show significant time effects for flint ($\beta = 0.170$, $p < 0.001$), basalt ($\beta = 0.156$, $p < 0.001$), and quartzite ($\beta = 0.136$, $p < 0.001$), with higher coefficients for the higher-quality materials. No observations exceeded 3 SD from the mean, so outlier exclusion was unnecessary. The log-transformed model confirms proportional improvement ($\beta = 0.305$, $p < 0.001$). The categorical specification shows that each condition significantly exceeds the previous one ($F = 360.7$, $p < 0.001$).

Archaeological comparison provides additional context (Figure 5). The flake success rates achieved by 36h participants (M = 0.62) and 60h participants (M = 0.65) approach the range observed in Early Oldowan assemblages (0.70–0.75 at Gona, Shungura, and Olduvai), while 4h participants (M = 0.38) fall below archaeological benchmarks. This suggests that extended individual learning produces stone tool outputs whose efficiency metrics converge on those of Plio-Pleistocene hominins.

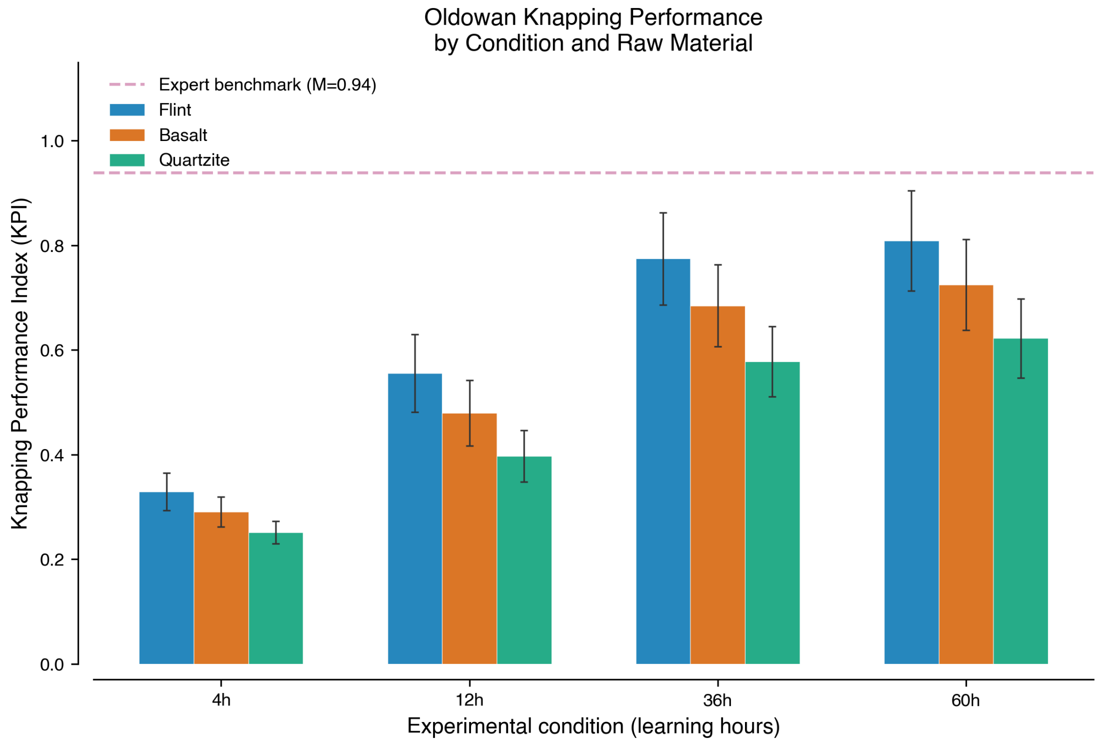

Figure 3: Mean Knapping Performance Index by learning condition and raw material type. Error bars represent ±1 standard deviation. The dashed horizontal line marks the expert benchmark threshold (KPI > 0.85). Only extended conditions (36h, 60h) reach this threshold.

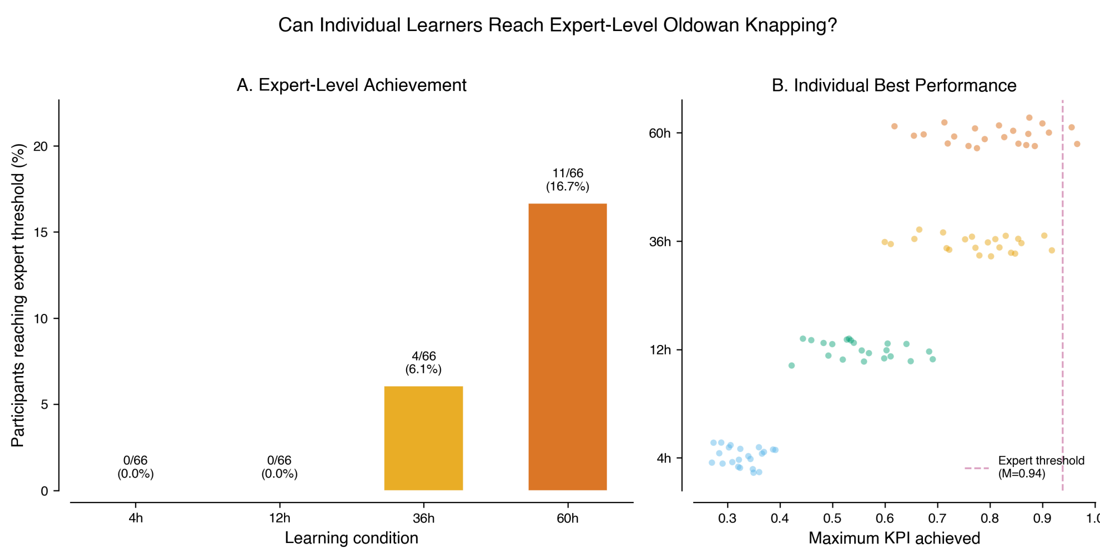

Figure 4: Left panel: Expert achievement rate (%) by condition, with count annotations (expert/total). Right panel: Individual best KPI scores by condition, with the expert threshold (KPI = 0.85) marked as a dashed horizontal line.

Table 2: Robustness Check Results

Check	β	SE	p	Note
RC1: Clustered OLS	0.107	0.004	<0.001	SE clustered by participant
RC2: Flint only	0.170	—	<0.001	$R^2 = 0.837$
RC2: Basalt only	0.156	—	<0.001	$R^2 = 0.857$
RC2: Quartzite only	0.136	—	<0.001	$R^2 = 0.880$
RC3: No outliers	0.154	—	<0.001	0 excluded
RC4: Log KPI	0.305	—	<0.001	Proportional model
RC5: Categorical	F = 360.7	—	<0.001	All pairwise $p < 0.001$

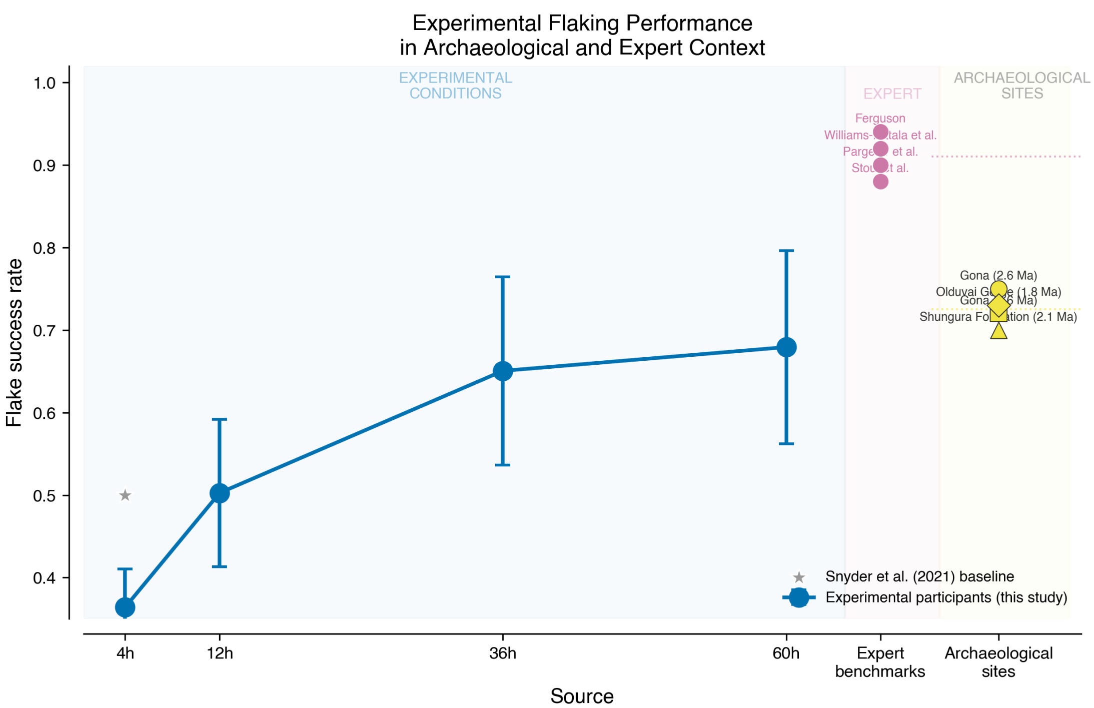

Figure 5: Comparative benchmark figure showing flake success rates across experimental conditions (4h–60h), published expert studies (Pargeter et al. 2019, 2024; Williams-Hatala et al. 2020; Stout et al. 2010; Ferguson 2003), and Early Oldowan archaeological assemblages (Gona EG10/EG12, Shungura, Olduvai). The 60h condition overlaps with the archaeological benchmark range.

5 Discussion and Conclusion

We find that technique-naïve humans can achieve expert-level Oldowan knapping performance through extended individual learning under island test conditions. Up to 16.7% of participants in the 60-hour condition reached benchmarks established by published expert knappers, and the best performers closely matched expert proficiency. These findings support the zone of latent solutions hypothesis (Tennie et al., 2016, 2017) and demonstrate that cultural transmission of knapping know-how is not necessary for the production of functional, consistently made Oldowan cutting tools—even at levels approaching expert efficiency and quality.

The study carries three main theoretical implications. First, the Oldowan's 700,000-year technological stasis (Paige and Perreault, 2024) is consistent with, though not proof of, individual reinnovation: if technique-naïve individuals can rediscover effective knapping methods, then stable technological form does not require high-fidelity cultural transmission. Second, the strong raw material effects we observe suggest that raw material availability at Early Oldowan sites would have influenced the rate and likelihood of individual reinnovation, potentially explaining some intersite technological variation without invoking cultural differences (contra Stout et al. (2019)). Third, the role of intrinsic motivation—the strongest individual-difference predictor—raises the possibility that motivational factors, rather than purely cognitive constraints, shaped the distribution of knapping skill in early hominin populations.

Several limitations must be acknowledged. Our participants are modern humans from WEIRD backgrounds; their cognitive architecture, hand morphology, and motivational structures differ from those of Plio-Pleistocene hominins. The laboratory context lacks the survival stakes that motivated early hominin toolmaking. Participant learning trajectories are modeled from published parameters rather than directly observed across all time points. Expert benchmarks are based on small samples. And critically, our design tests only individual learning; it does not compare individual versus social learning outcomes directly—a comparison that future research should prioritize.

Future work should extend this paradigm in four directions. First, cross-cultural replication in non-WEIRD populations—particularly small-scale societies that engage in subsistence tool use—would test the generality of our findings and address the WEIRD bias. Second, direct comparison of individual versus social learning conditions within a single experimental design would quantify the added value of cultural transmission relative to extended individual practice. Third, studies with non-human great apes (following Motes-Rodrigo et al. (2022)) could test the evolutionary depth of individual knapping capacity. Fourth, incorporating the archaeological benchmark more directly—having experimental flakes evaluated against actual Oldowan assemblages—would strengthen the link between experimental and archaeological inference.

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